

# OBSERVATION

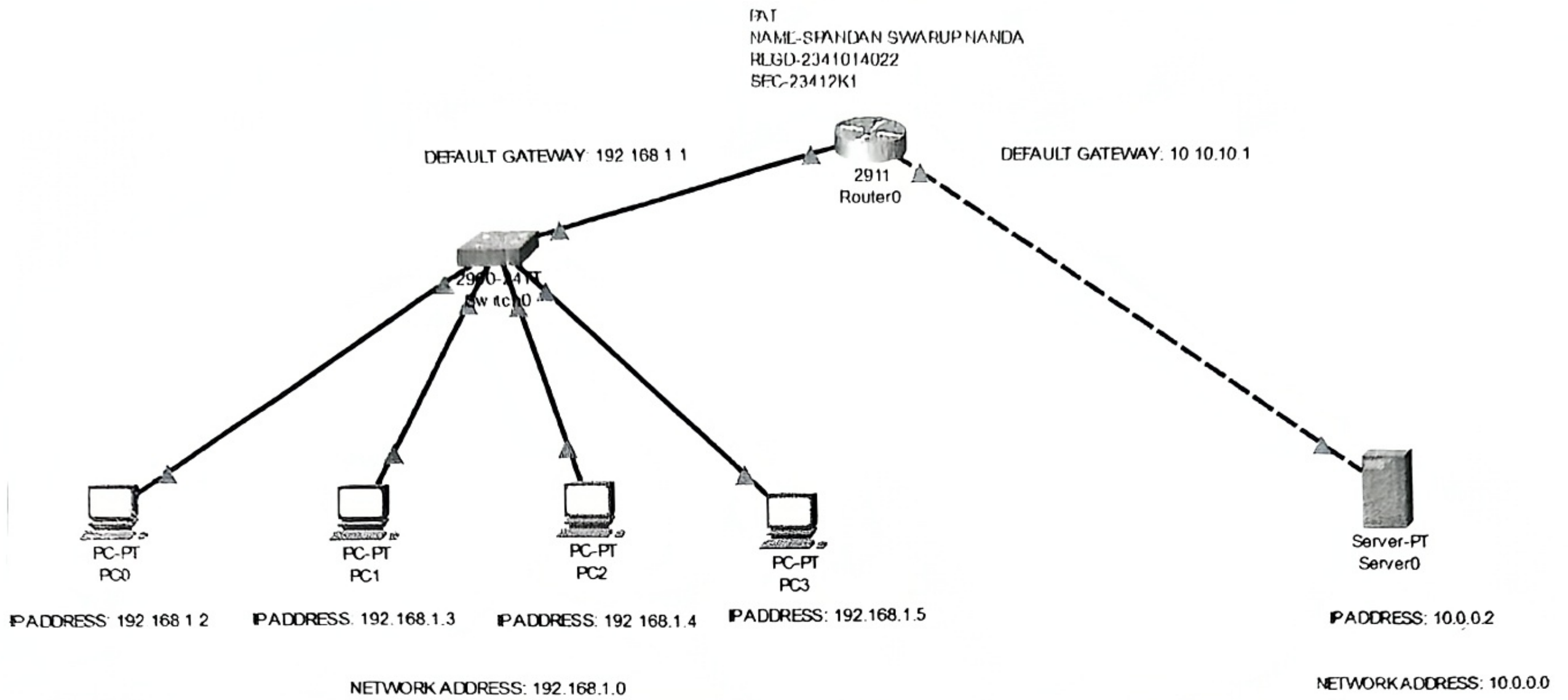


Fig. Network topology for PAT Configuration

The diagram shows a private LAN connected to a public network through a router using Port Address Translation (PAT).

Fig. Router configuration for PAT using CLI  
 Router interfaces are configured as NAT inside and NAT outside, and PAT is enabled using access control list and overload command.

```

Router0
Physical Config CLI Attributes
IOS Command Line Interface

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#interface GigabitEthernet0/1
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router(config-if)#exit
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/1
Router(config-if)#ip address 10.10.10.1 255.0.0.0
Router(config-if)#ip address 10.10.10.1 255.0.0.0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/2
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/1
Router(config-if)#
Router(config-if)#interface gig 0/0
Router(config-if)#ip nat inside
Router(config-if)#exit
Router(config)#interface gig 0/1
Router(config-if)#ip nat outside
Router(config-if)#exit
Router(config)#access-list 1 permit 192.168.1.0 0.0.0.255
Router(config)#ip nat inside source list 1 interface gig 0/1 overload

% Invalid input detected at '^' marker.

Router(config)#ip nat inside source list 1 interface gig 0/1 overload
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip nat translations
Router#show ip nat translations
  
```



Router0

Physical
Config
CLI
Attributes

IOS Command Line Interface

```

Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/2
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/1
Router(config-if)#
Router(config-if)#interface gig 0/0
Router(config-if)#ip nat inside
Router(config-if)#exit
Router(config)#interface gig 0/1
Router(config-if)#ip nat outside
Router(config-if)#exit
Router(config)#access-list 1 permit 192.168.1.0 0.0.0.255
Router(config)#ip nat inside source list 1 interface gig 0/1 overload
                                     ^
% Invalid input detected at '^' marker.

Router(config)#ip nat inside source list 1 interface gig 0/1 overload
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip nat translations
Router#show ip nat translations
Pro  Inside global      Inside local      Outside local      Outside global
icmp 10.10.10.1:1024     192.168.1.3:1     10.0.0.2:1         10.0.0.2:1024
icmp 10.10.10.1:1025     192.168.1.3:2     10.0.0.2:2         10.0.0.2:1025
icmp 10.10.10.1:1026     192.168.1.3:3     10.0.0.2:3         10.0.0.2:1026
icmp 10.10.10.1:1027     192.168.1.3:4     10.0.0.2:4         10.0.0.2:1027
icmp 10.10.10.1:1028     192.168.1.4:2     10.0.0.2:2         10.0.0.2:1028
icmp 10.10.10.1:1029     192.168.1.4:3     10.0.0.2:3         10.0.0.2:1029
icmp 10.10.10.1:1030     192.168.1.4:4     10.0.0.2:4         10.0.0.2:1030
icmp 10.10.10.1:1031     192.168.1.5:3     10.0.0.2:3         10.0.0.2:1031
icmp 10.10.10.1:1032     192.168.1.5:4     10.0.0.2:4         10.0.0.2:1032
icmp 10.10.10.1:1033     192.168.1.5:5     10.0.0.2:5         10.0.0.2:1033
icmp 10.10.10.1:1       192.168.1.2:1     10.0.0.2:1         10.0.0.2:1
icmp 10.10.10.1:2       192.168.1.2:2     10.0.0.2:2         10.0.0.2:2
icmp 10.10.10.1:3       192.168.1.2:3     10.0.0.2:3         10.0.0.2:3
icmp 10.10.10.1:4       192.168.1.2:4     10.0.0.2:4         10.0.0.2:4
icmp 10.10.10.1:5       192.168.1.4:5     10.0.0.2:5         10.0.0.2:5
icmp 10.10.10.1:6       192.168.1.5:6     10.0.0.2:6         10.0.0.2:6

Router#

```

**Fig. NAT Translation Table Output**

The NAT table displays multiple private IP addresses translated to a single public IP using different port numbers, confirming PAT operation.

• ANALYSIS

Network Topology Overview

- One Router (2911) connects two different networks.
- Inside LAN: 192.168.1.0/24 with multiple PCs via a switch.
- Outside Network: 10.0.0.0/8 with a server.
- Router acts as the gateway and NAT device.



## IP Addressing and Gateway

- PCs assigned IPs: 192.168.1.2 - 192.168.1.5
- Inside default gateway: 192.168.1.1
- Outside interface IP: 10.10.10.1
- Server IP: 10.0.0.2

## Interface Configuration

- GigabitEthernet0/0 → configured as NAT inside
- GigabitEthernet0/1 → configured as NAT outside
- Both interfaces are enabled (no shutdown)

## Access Control List (ACL)

- ACL permits traffic from 192.168.1.0/24
- This ACL identifies inside local address for NAT

## PAT (NAT Overload) Operation

- Command used:  
ip nat inside source list 1 interface g0/1 overload.
- Multiple internal hosts share one public IP (10.10.10.1)
- Different port numbers are used to distinguish sessions.

## NAT Translation Table Observation

- show ip nat translations confirms PAT working.
- Multiple inside local IPs (192.168.1.n) mapped to:
  - One inside global IP (10.10.10.1)
- ICMP entries show successful translations during ping.

## Connectivity Verification

- PCs can successfully ping the server
- NAT entries dynamically created during traffic flow.
- Confirms correct PAT configuration and security.

## CONCLUSION

This experiment successfully demonstrated the concept and working of Network Address Translation (NAT) in IP networking. Static NAT, Dynamic NAT, and PAT were configured using both CLI and Config modes, enabling communication between private and public networks. The NAT translation table was analyzed to verify correct IP address mapping, confirming effective address conservation and secure internal network communication as intended.



## • EXERCISES

- Q1) What is the purpose of NAT in IP networking? Why is NAT used in IPv4 networks?
- Q2) What is Inside Local and Inside Global addresses?
- Q3) What are the differences between Static NAT, Dynamic NAT, and PAT?
- Q4) How does PAT allow multiple devices to share a single public IP address?
- Q5) What command is used to view active NAT translations? How do you verify that NAT translations are working correctly in Cisco Packet Tracer?
- Q1) NAT translates private IPv4 addresses into public IPv4 addresses, allowing private networks to access the internet. It is used in IPv4 networks to conserve public IP addresses and hide internal network structure.
- Q2) Inside Local Address: The private IPv4 address assigned to an internal device (e.g., 192.168.1.10).  
Inside Global Address: The public IPv4 address that represents the internal device on the internet.
- Q3) Static NAT: One-to-one mapping between private and public IPs.  
Dynamic NAT: Many-to-many mapping using a pool of public IPs.  
PAT: Many-to-one mapping using port numbers.  
eg: PAT allows multiple private hosts to share a single public IP
- Q4) PAT uses unique source port numbers along with the same public IP address to distinguish multiple internal devices, allowing simultaneous internet access.
- Q5) Command: show ip nat translations  
Verification: Successful ping from private to public network and correct NAT entries in the translation table confirm NAT operation.